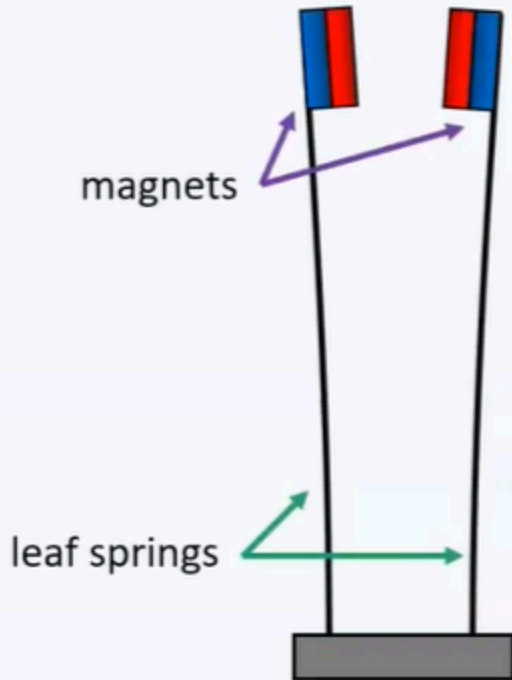
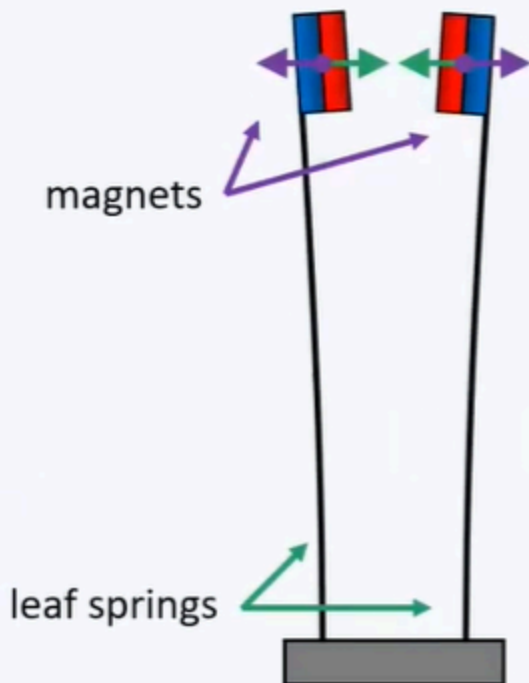


Problem content



Secure the lower ends of two identical leaf springs to a non-magnetic base and attach magnets to the upper ends such that they repel and are free to move. Investigate how the movement of the springs depends on relevant parameters.

Qualitative explanation



Leaf springs

Acts as *springs*, pushing disturbed system back to equilibrium

We will model leaf springs as beams using **Euler-Bernoulli equations**

Magnets

Allow for interaction between beams (i.e. coupling)

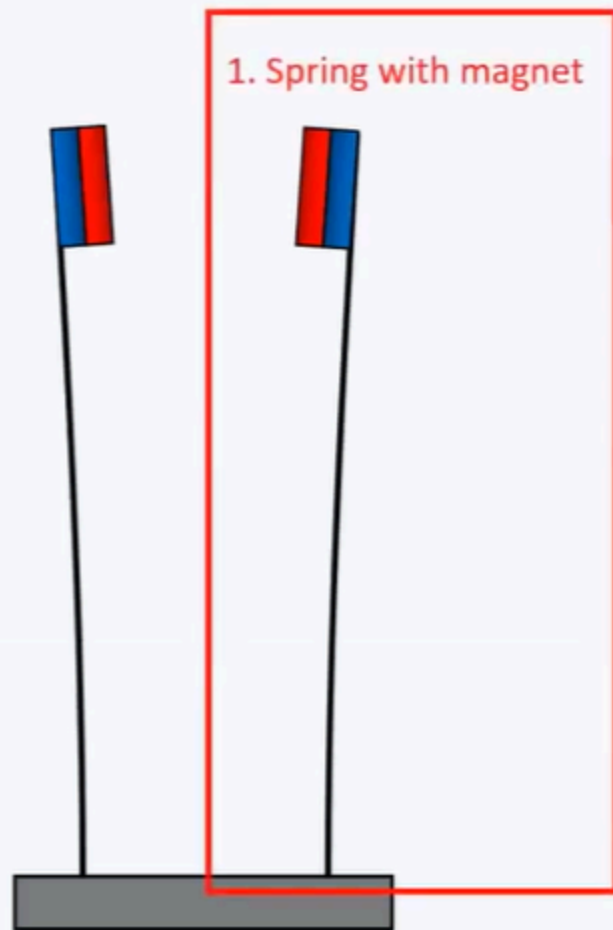
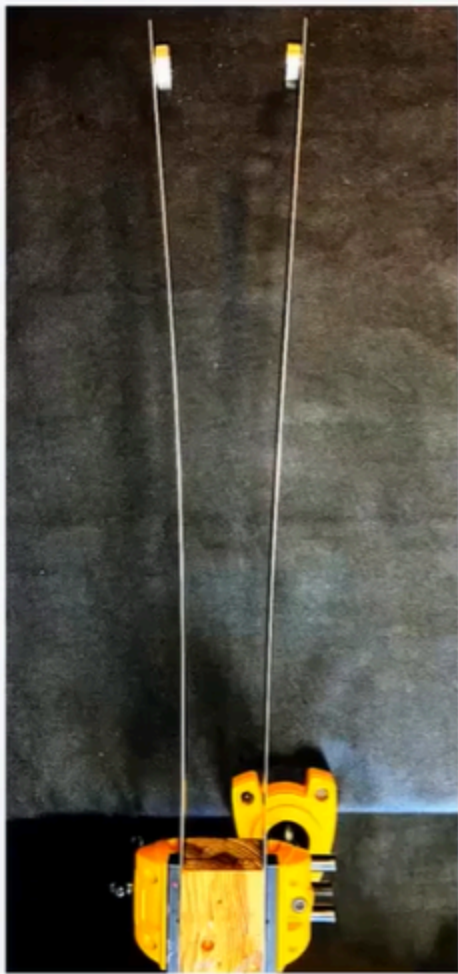
We will use **dipole approximation** with magnetic dipole moment fitted using full cylindrical magnet model

Gravity

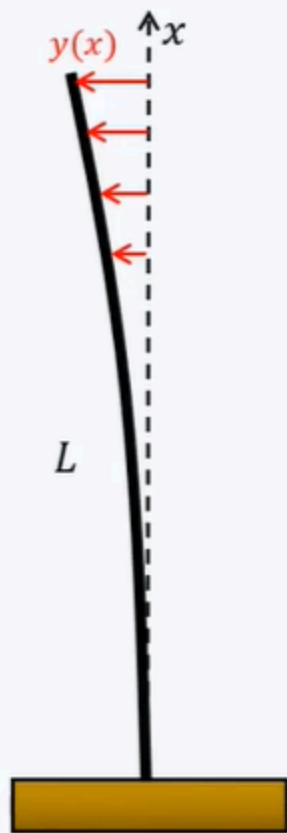
Results in additional forces and torques acting on the magnets

We will eliminate the effect of gravity by investigate system **oriented horizontally** – effect of gravity can be neglected

Setup diagram



Dynamics of magnet and spring



Euler-Bernoulli equation

$$\mu \frac{\partial^2 y}{\partial t^2} = -EI \frac{\partial^4 y}{\partial x^4}$$

Boundary conditions

I		$y(0) = 0$
II		$\frac{\partial y(0)}{\partial x} = 0$
III		$\frac{\partial^2 y(l)}{\partial x^2} = 0$
IV		$EI \frac{\partial^3 y(l)}{\partial x^3} = 0$

Vibration eigenmodes

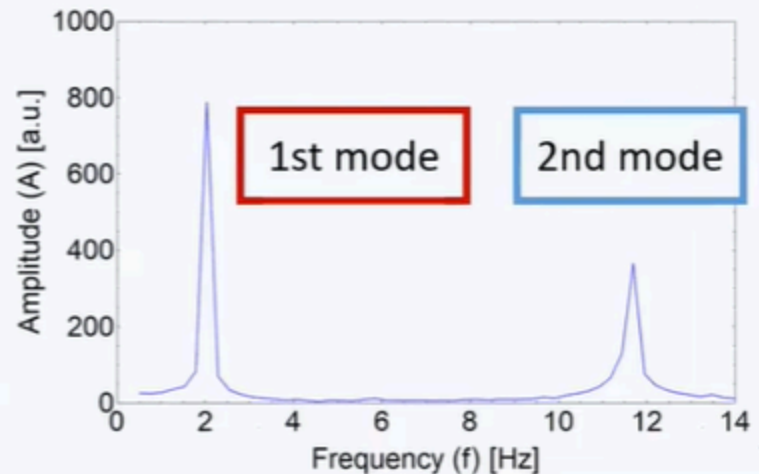
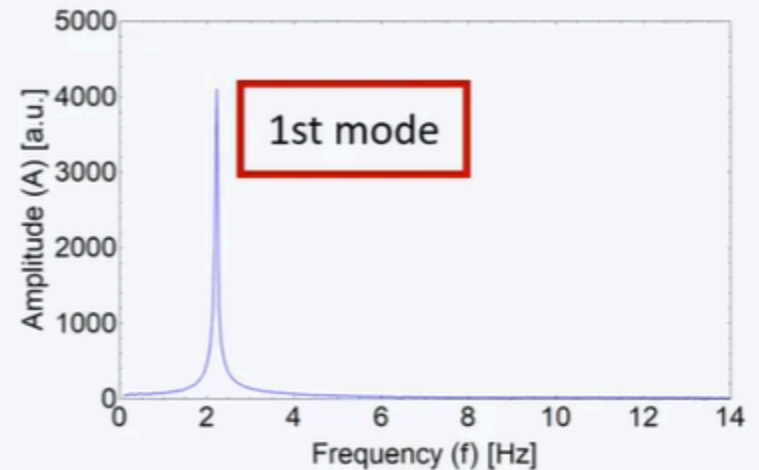
$$\cos \beta \cosh \beta + 1 = 0,$$

with

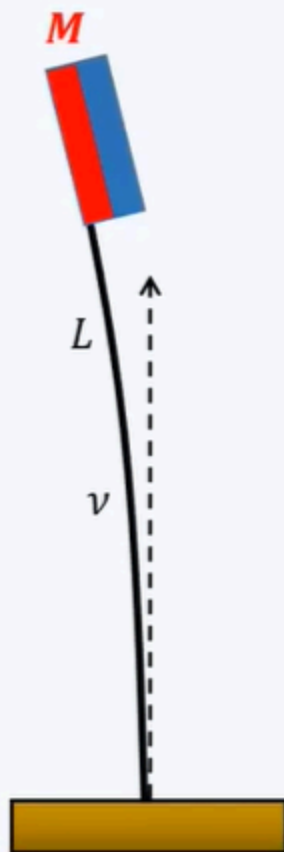
$$\beta^4 = \frac{L^4 \mu \omega^2}{EI}$$

Where: EI – bending stiffness, x – distance from base, y – deflection, μ – linear density, ω – angular frequency, t – time, l – length of the beam, L – length of the spring

Higher modes of oscillation



Dynamics of magnet and spring



Euler-Bernoulli equation

$$\mu \frac{\partial^2 y}{\partial t^2} = -EI \frac{\partial^4 y}{\partial x^4}$$

Boundary conditions

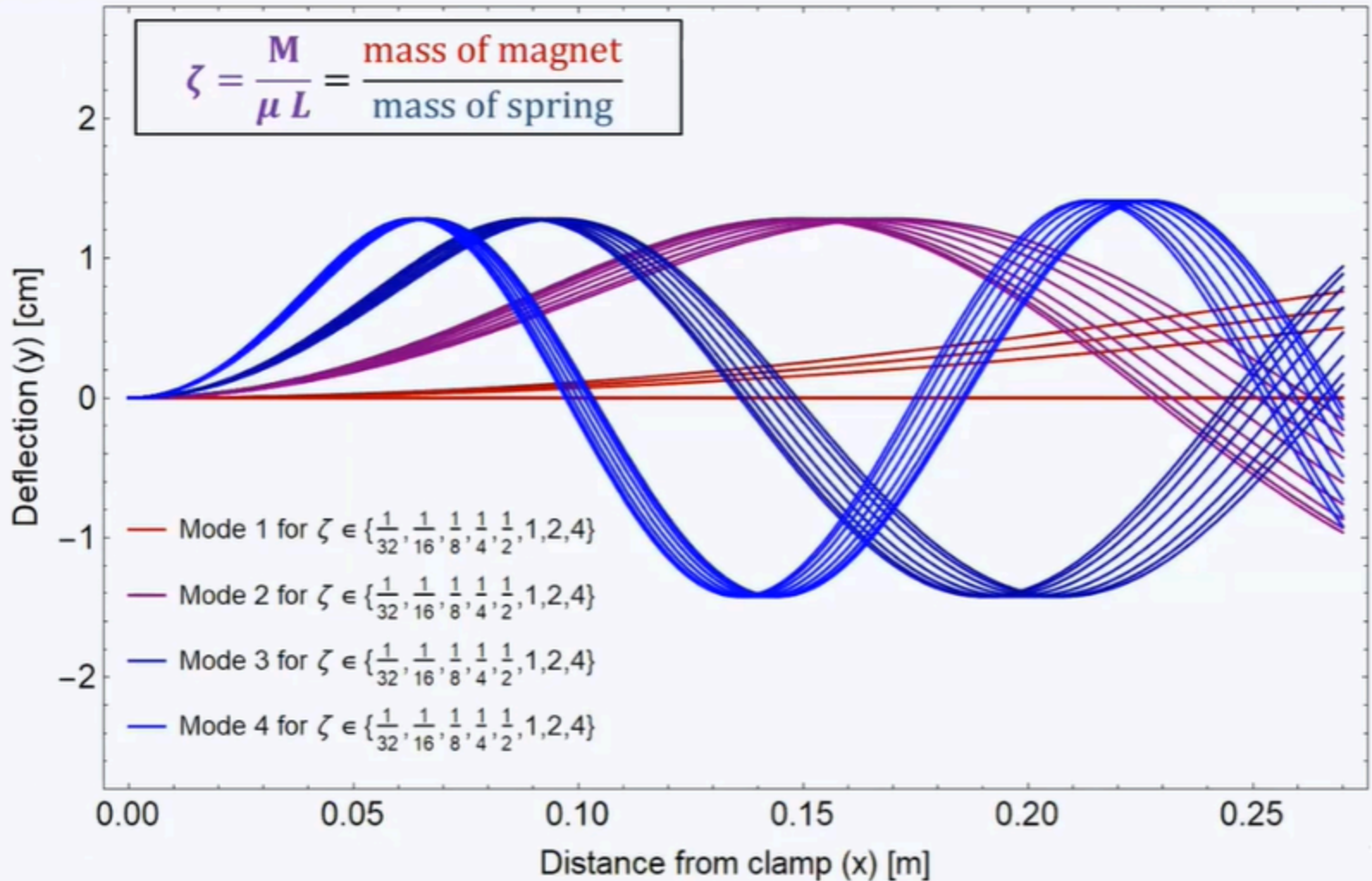
I		$y(0) = 0$
II		$\frac{\partial y(0)}{\partial x} = 0$
III		$\frac{\partial^2 y(l)}{\partial x^2} = 0$
IV		$EI \frac{\partial^3 y(l)}{\partial x^3} - M \frac{\partial^2 y(l)}{\partial t^2} = 0$

Vibration eigenmodes

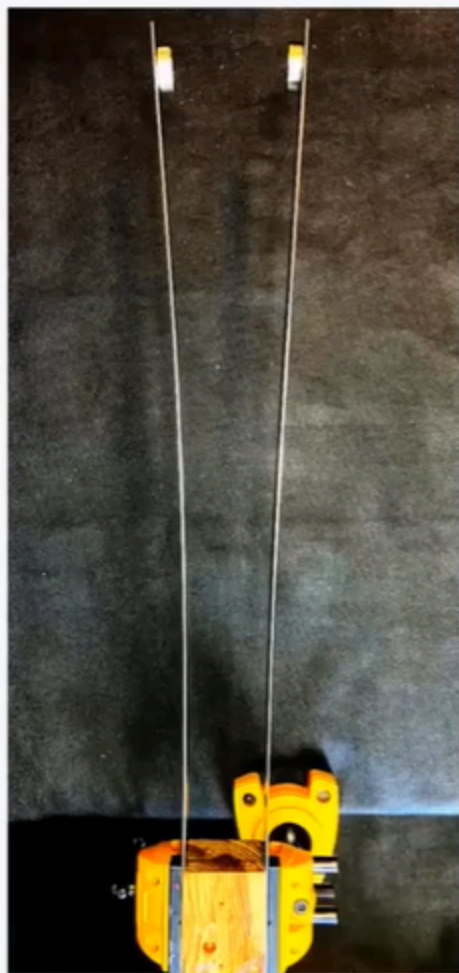
$$\beta \frac{M}{\mu L} (\cos \beta \sinh \beta - \cosh \beta \sin \beta) - \cosh \beta \cos \beta + 1 = 0 \quad \beta^4 = \frac{L^3 (\mu L) \omega^2}{EI}$$

Where: EI – bending stiffness, x – position along the beam, y – deflection, μ – linear density, ω – angular frequency, t – time, l – length of beam, L – length of spring, M – mass of spring

Model of beam with magnet



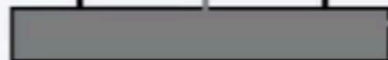
Setup diagram



2. Magnet



1. Spring with magnet



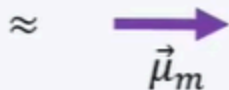
Magnetic field from magnet



Magnet



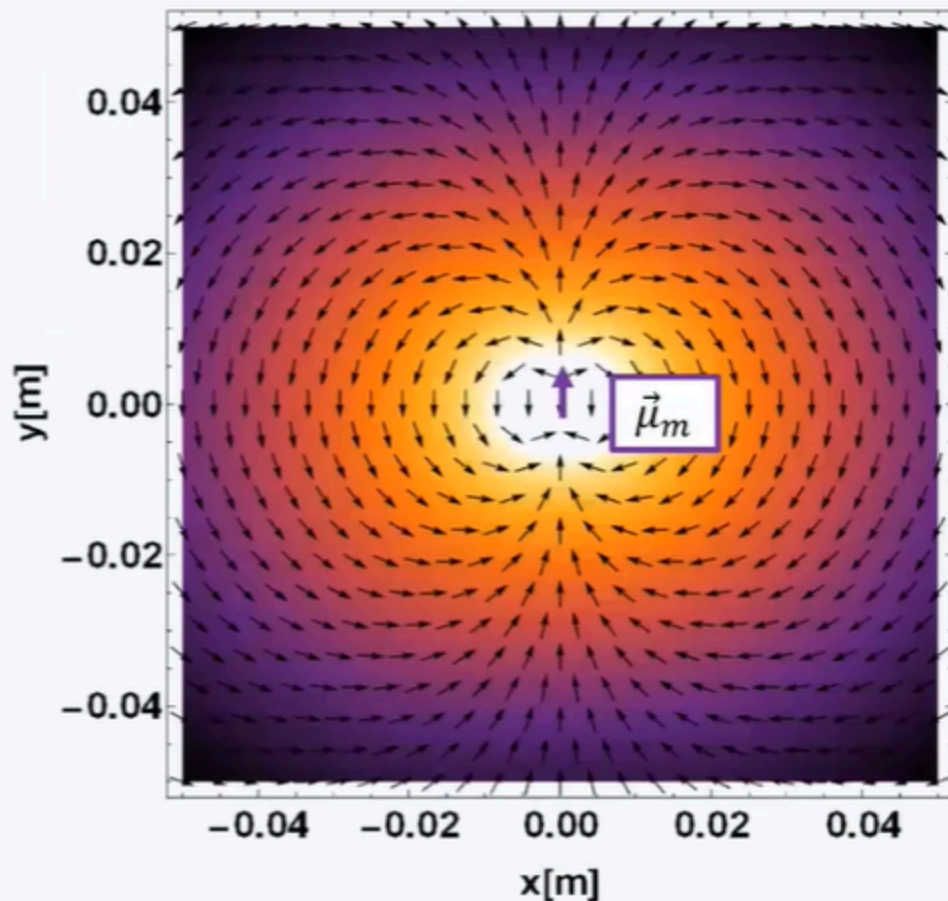
Magnetic dipole



Magnetic field from dipole

$$\vec{B}(\vec{r}) = \frac{\mu_0}{4\pi} \left(\frac{3\vec{r}(\vec{\mu}_m \cdot \vec{r})}{r^5} - \frac{\vec{\mu}_m}{r^3} \right)$$

$$\vec{B} = -\frac{\mu_0 \mu_m}{4\pi r^3} \begin{pmatrix} -3 \sin \theta \cos \theta \\ 3 \cos^2 \theta - 1 \\ 0 \end{pmatrix}$$



Where: $\vec{B}$ – magnetic field, μ_0 – vacuum permeability, $\vec{\mu}_m$ – magnetic dipole moment, $\vec{r}$ – distance from the dipole, θ – angle from y axis

Interaction between magnets



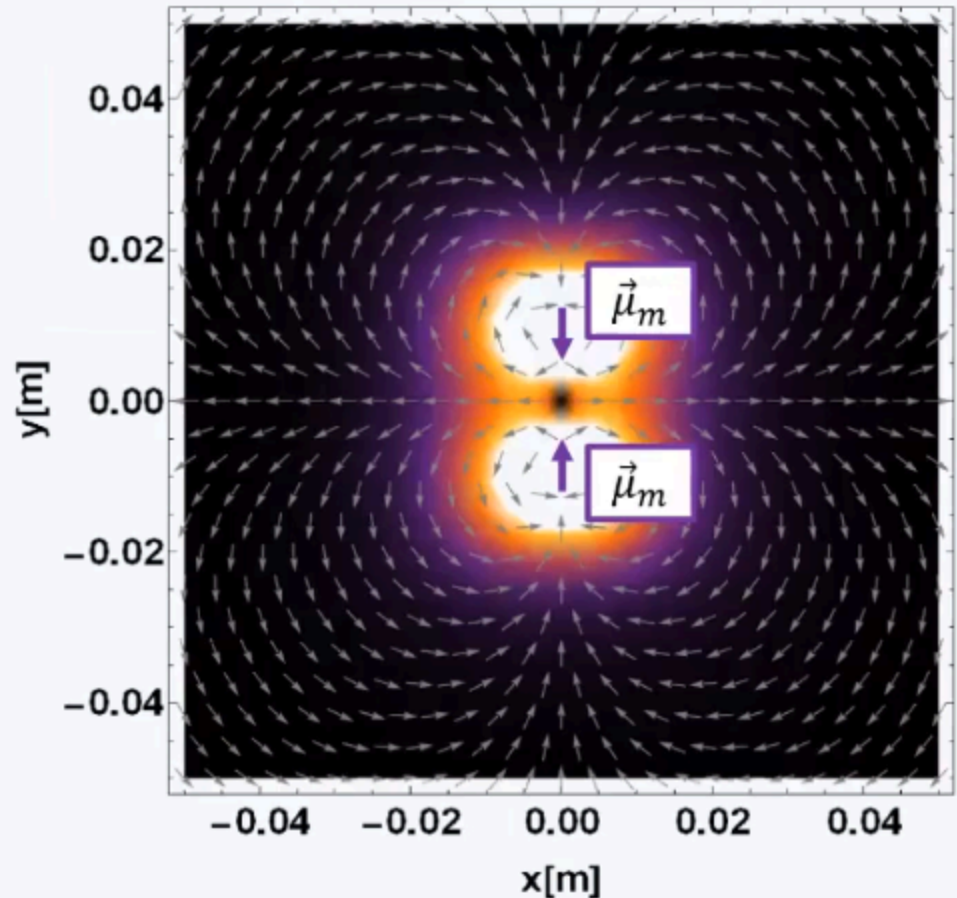
Force acting on dipole:

$$\vec{F}_m = \nabla(\vec{\mu}_m \cdot \vec{B})$$

Motion occurs in one axis

For paralel, identical magnets:

$$F_m = \frac{3\mu_0 \mu_m^2}{2\pi r^4}$$



Where: $\vec{B}$ – magnetic field, μ_0 – permeability of free space, $\vec{\mu}_m$ – magnetic dipole moment, r – distance between magnets, $\vec{F}_m$ – force acting on a dipole

Shift of equilibrium point



Magnetic force

$$F_m = \frac{3 \mu_0 \mu_m^2}{2 \pi r^4}$$

Elastic force

$$F_e = \frac{3 EI}{L^3} x$$

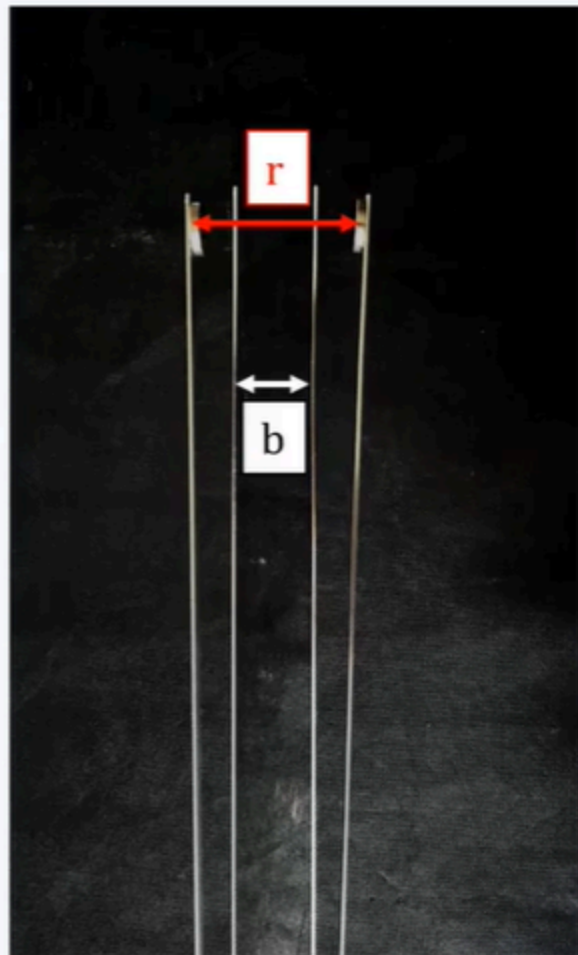
Equilibrium: $F_m = F_e$

Quintic equation for equilibrium point

$$r_0^4 (r_0 - b) = \frac{\mu_0 \mu_m^2 L^3}{\pi EI}$$

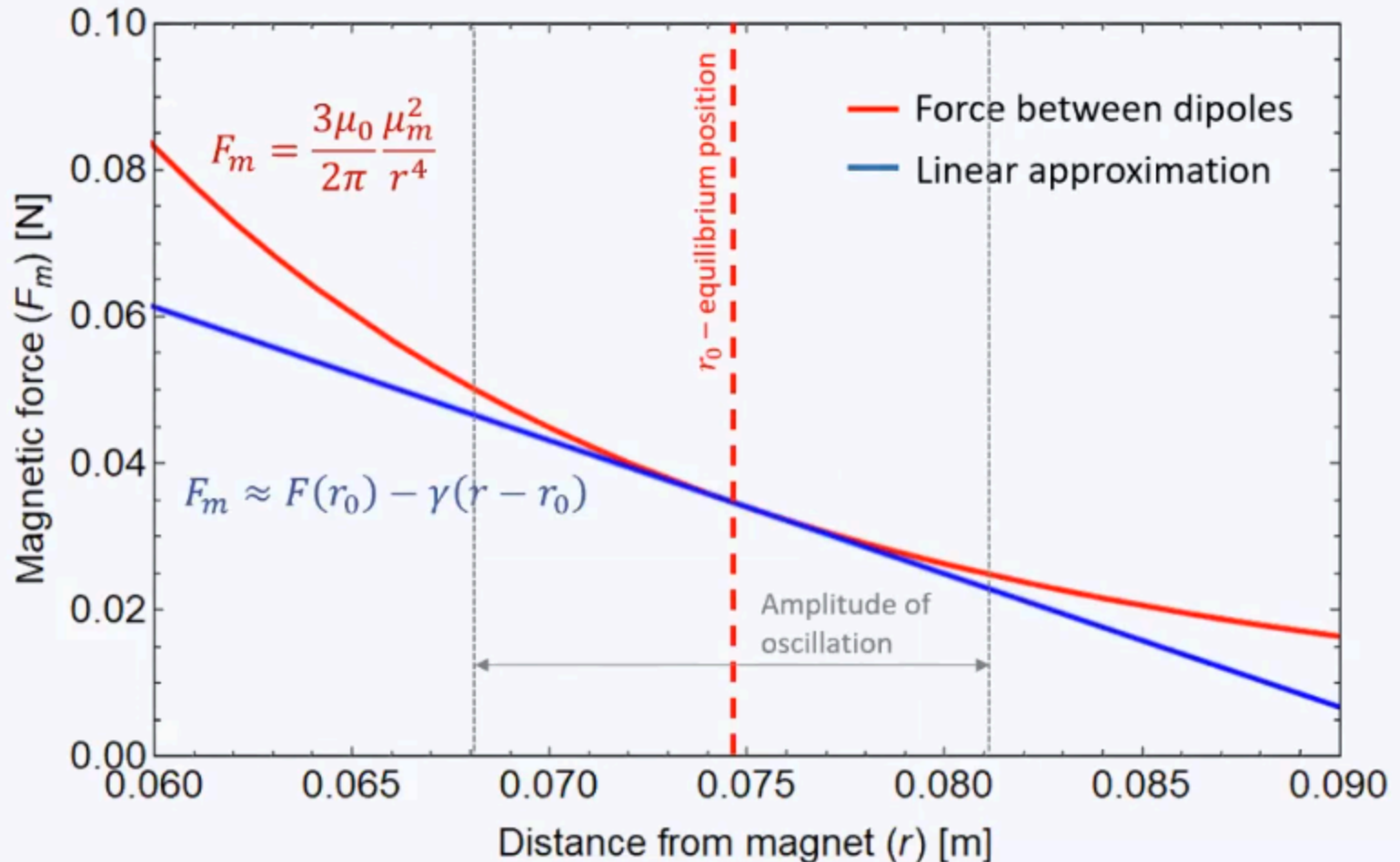
Approximation around the equilibrium point

$$F_m = F(r_0) - \frac{6 \mu_0 \mu_m^2}{\pi r_0^5} (r - r_0) + O((r - r_0)^2)$$

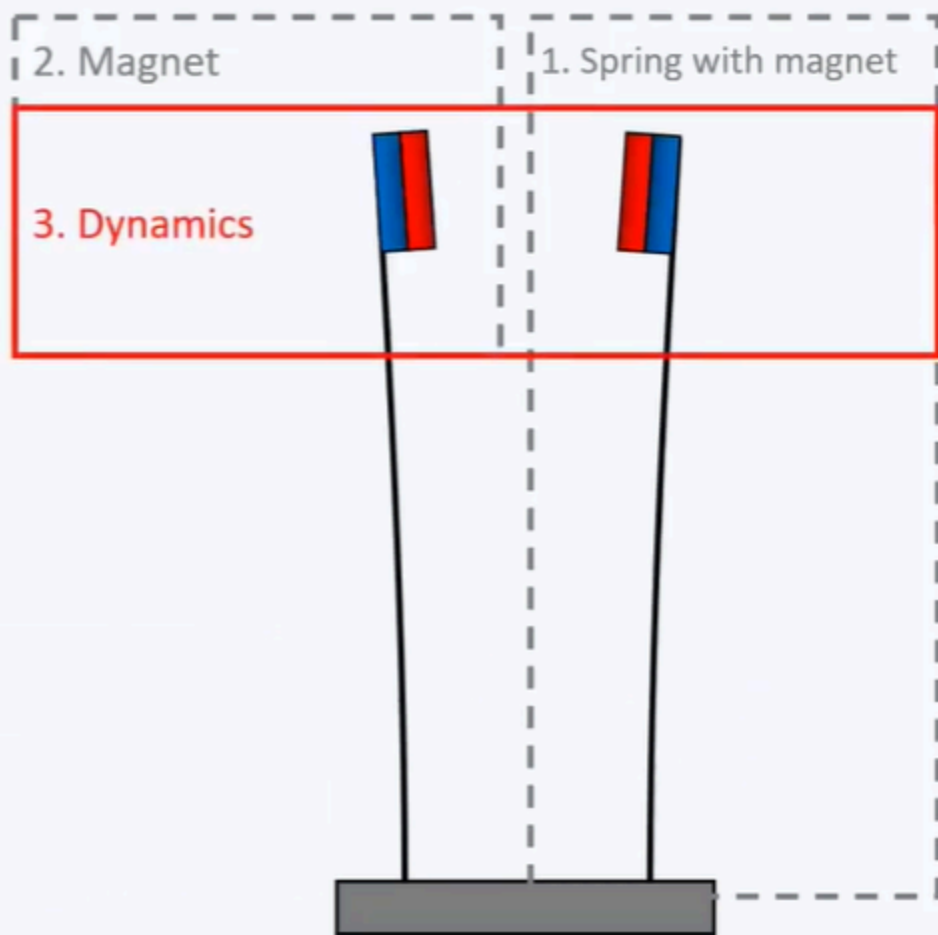
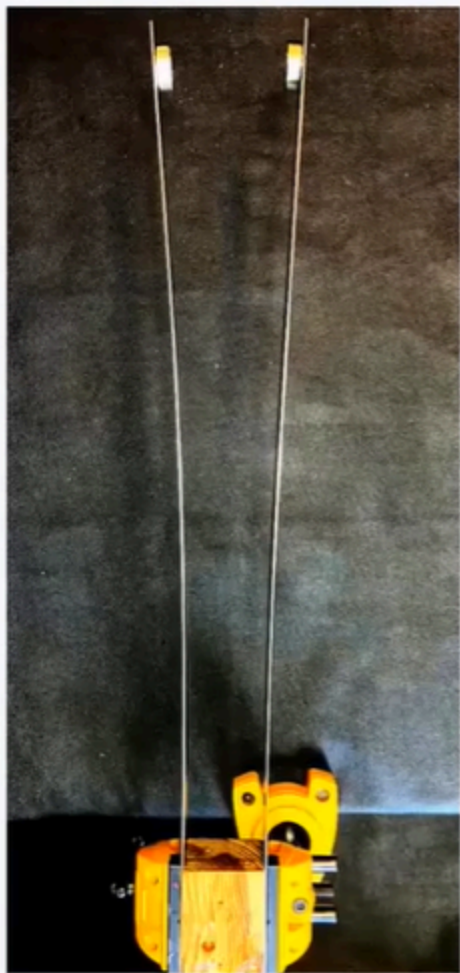


Where: μ_0 – vacuum permeability, μ_m – magnetic dipole moment, r – distance between magnets, b – width of base, L – length of spring, EI – bending stiffness

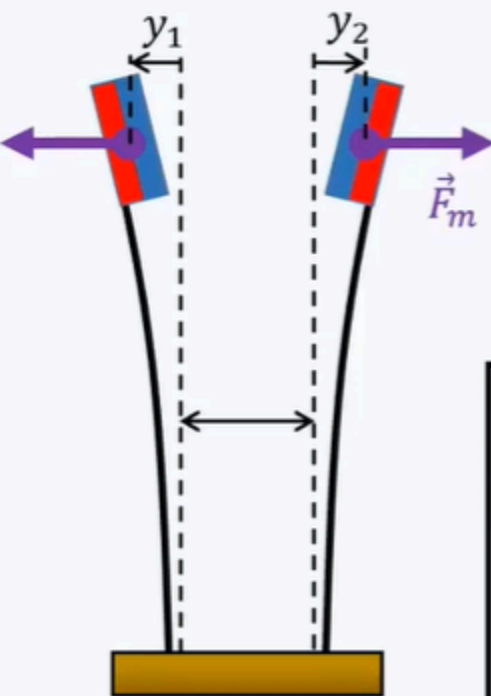
Linear approximation



Setup diagram



Equations of motion



Euler-Bernoulli equation

$$\mu \frac{\partial^2 y}{\partial t^2} = EI \frac{\partial^4 y}{\partial x^4}$$



Boundary
conditions

I



$$y(0) = 0$$

II



$$\frac{\partial y(0)}{\partial x} = 0$$

III



$$\frac{\partial^2 y(l)}{\partial x^2} = 0$$

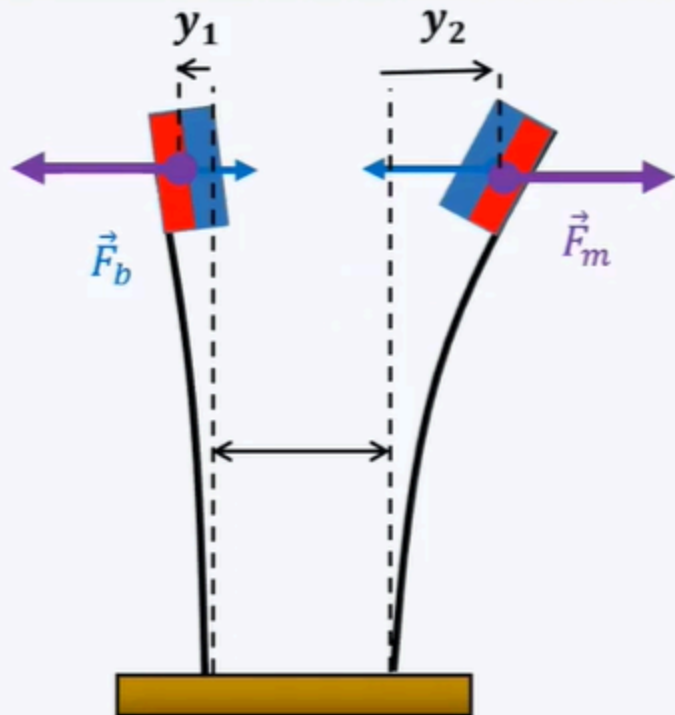
IV



$$EI \frac{\partial^3 y(l)}{\partial x^3} - M \frac{\partial^2 y(l)}{\partial t^2} + \text{magnets} = 0$$

Where μ_0 – permeability of free space, d – distance between magnets, EI – bending stiffness, L – length of spring, m – mass of magnet, M – mass of spring

Equations of motion of free end



This system has two degrees of freedom:
 $\{y_1, y_2\}$

Equations of motion at free end:

$$\begin{cases} M \frac{\partial^2 y_1(l)}{\partial t^2} = EI \frac{\partial^3 y_1(l)}{\partial x^3} + \frac{\gamma}{2} (-y_1 + y_2) \\ M \frac{\partial^2 y_2(l)}{\partial t^2} = EI \frac{\partial^3 y_2(l)}{\partial x^3} - \frac{\gamma}{2} (-y_1 + y_2) \end{cases}$$

Inertia

Shear force

Magnetic force

$$\eta = y_1 + y_2$$

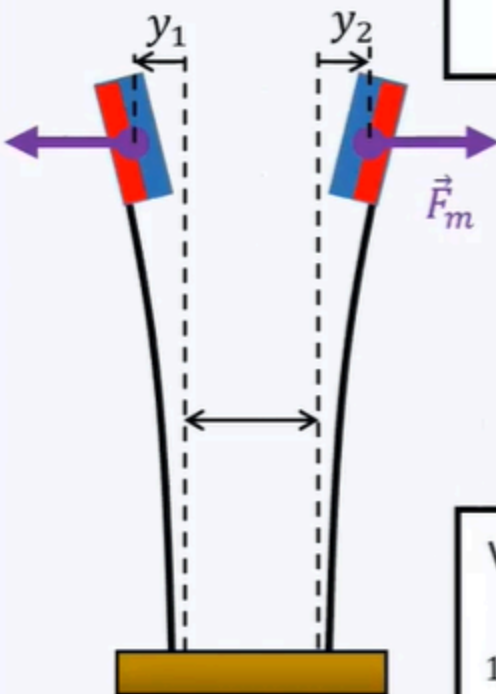
$$M \frac{\partial^2 \eta}{\partial t^2} = EI \frac{\partial^3 \eta}{\partial x^3}$$

$$\xi = y_1 - y_2$$

$$M \frac{\partial^2 \xi}{\partial t^2} = EI \frac{\partial^3 \xi}{\partial x^3} + \gamma \xi$$

Where: y_1, y_2 – degrees of freedom, L – length of spring, EI – bending stiffness, γ – linear magnetization coefficient, t – time, x – distance from base, M – mass of spring, l – length of beam

Equations of motion



Euler-Bernoulli equation

$$\mu \frac{\partial^2 \xi}{\partial t^2} = EI \frac{\partial^4 \xi}{\partial x^4}$$

Boundary conditions

I		$\xi(0) = 0$
II		$\frac{\partial \xi(0)}{\partial x} = 0$
III		$\frac{\partial^2 \xi(l)}{\partial x^2} = 0$
IV		$EI \frac{\partial^3 \xi(l)}{\partial x^3} - M \frac{\partial^2 \xi(l)}{\partial t^2} + \gamma \xi = 0$

Vibration Eigenmodes

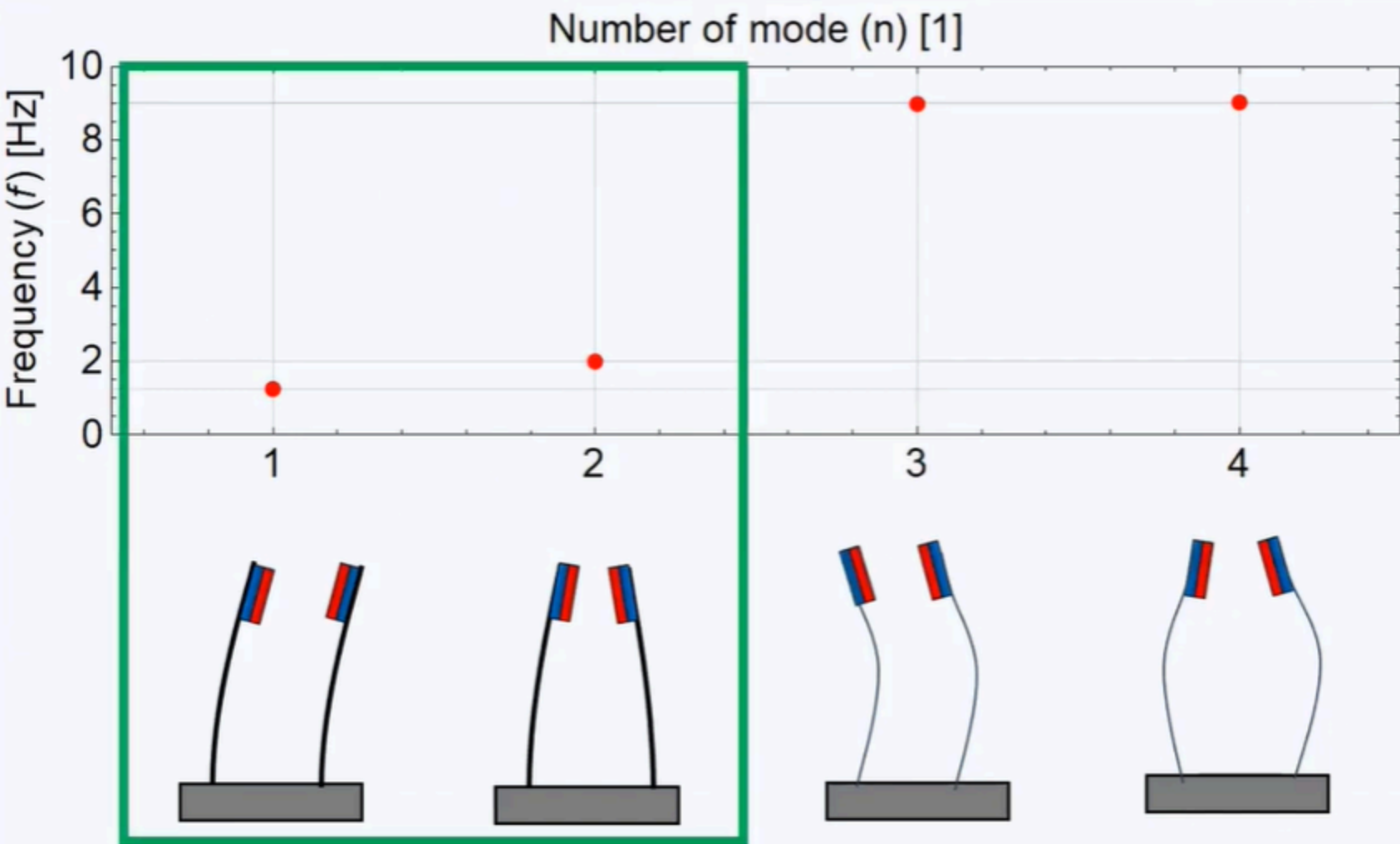
$$1 + \cosh \beta \left(\cos \beta - \left(h - \frac{M}{\mu L} \beta \right) \sin \beta \right) - \left(h - \frac{M}{\mu L} \beta \right) \sinh \beta \cos \beta = 0$$

$$\beta^4 = \frac{L^3 (\mu L) \omega^2}{EI}$$

$$h = \frac{2\gamma}{EI^{\frac{1}{4}} \mu^{\frac{3}{4}} \omega^{\frac{3}{2}}}$$

Where μ_0 – permeability of free space, d – distance between magnets, EI – bending stiffness, L – length of spring, m – mass of magnet, M – mass of spring

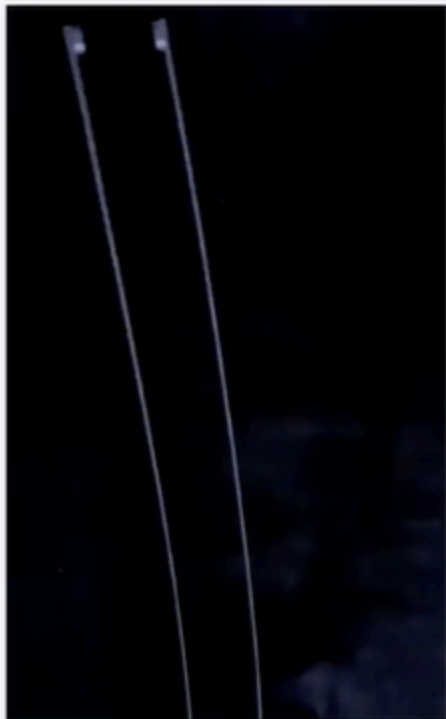
Frequency of the first modes



Superposition of modes



Mode 1

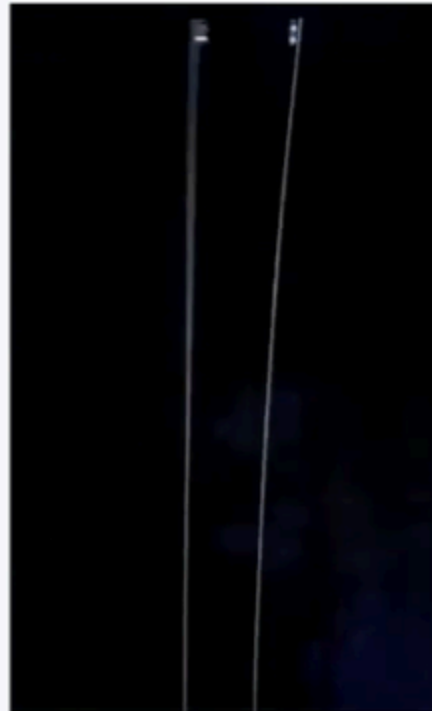


Mode 2



Solution

+

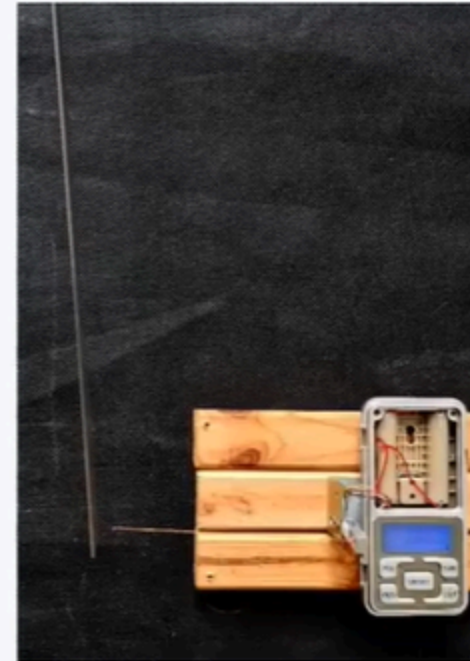
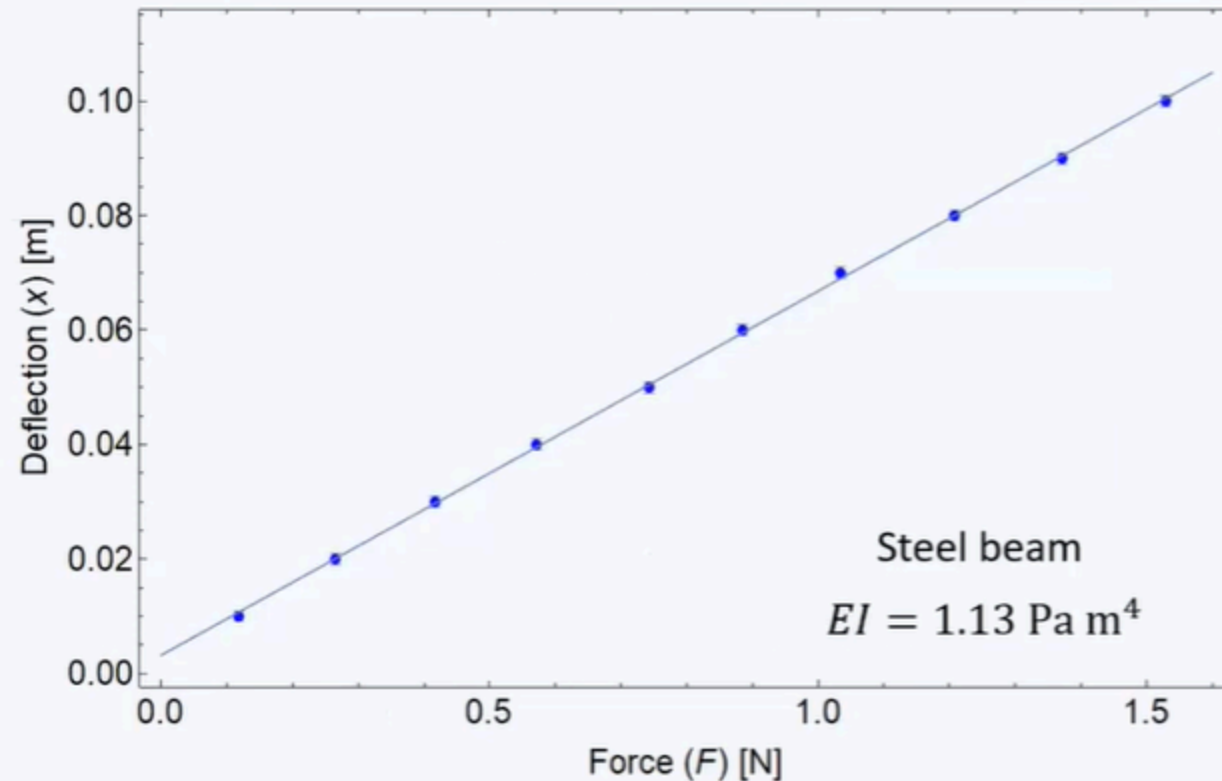


Measuring EI of all beams



Elastic force

$$F_e = \frac{3EI}{L^3} x$$



Plastic	0.40 Pa m^4
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Steel	1.13 Pa m^4
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Aluminum	0.34 Pa m^4
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Where: L – length of the beam, EI – bending stiffness, x – distance from base;

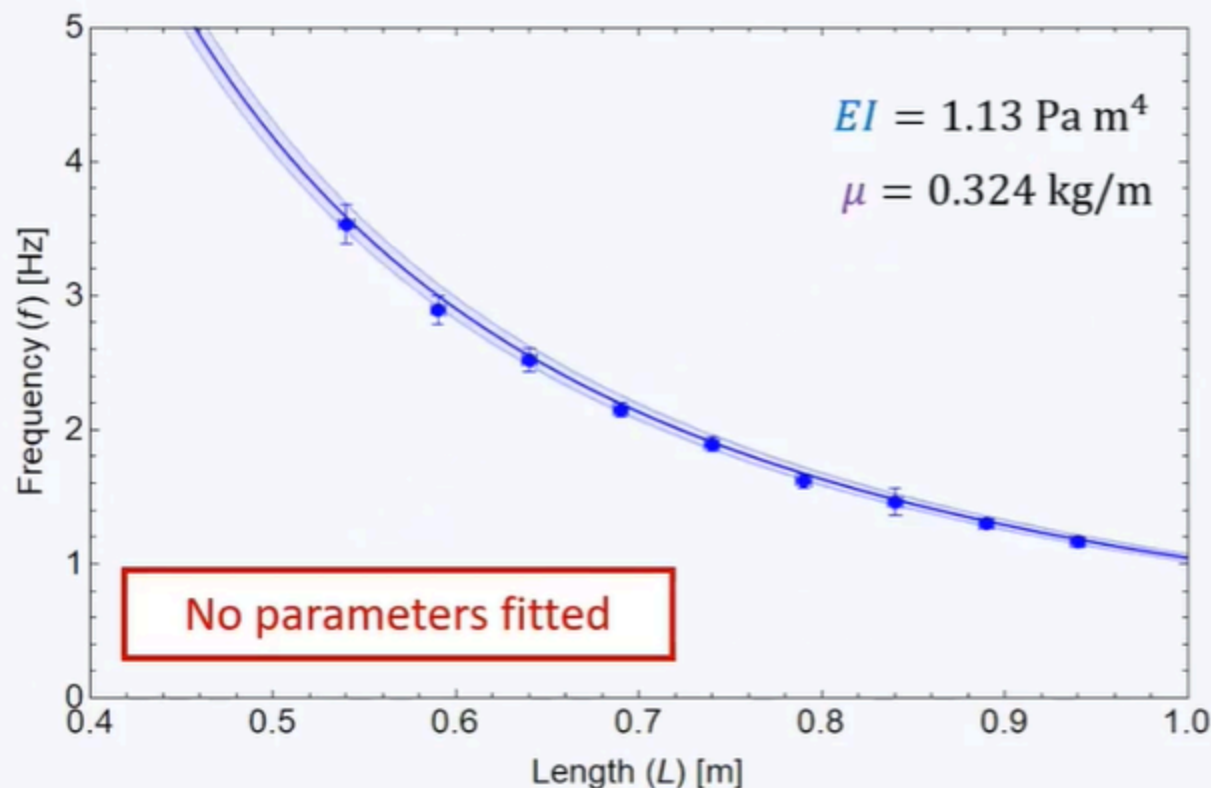
Free beam (Cantilever)



Vibration eigenmodes:

$$\cos \alpha \cosh \alpha + 1 = 0$$

$$\alpha^4 = L^4 \mu \omega^2 / EI$$



Where: L – length of the beam, μ – linear density, EI – bending stiffness, ω – angular frequency;